Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/31/2006
PI Name:	Theofanous, Theofanis G. Ph.D.		
Project Title:	The Limits of Coolability in Thermal Systems for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	10/01/2001	**End Date:**	11/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dinh, Truc-Nam (University of California Santa Barbara)		
Grant/Contract No.:	NAG3-2761		
Performance Goal No.:			
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Task Description:	This project is an integrative effort addressing practical aspects of boiling in the HEDS context. The work is a continuation and extension of a project that addressed coolability limits in pool boiling carried out over the previous funding cycle. A key result of this previous work was the identification of a scales separation phenomenon, that effectively "isolates" the microhydrodynamics on the heater surface from the chaotic two-phase flow motions away from it. This separation is effected by a persistent very high void fraction region (like a vapor "blanket"), and it is of immense significance in allowing to focus the physics of a long-standing seemingly intractable problem. Experiments and computations in this project will be used to examine applicability of scales separation as an organizing principle in convective boiling and boiling in fractional gravities. Task Objective: 1. Develop a basic understanding for the fine structure of the extended microlayer, as revealed through our "scale separation" idea and the dynamic thermal patterns on a nanofilm heater. 2. Define appropriate, measurable surface properties relevant to the rupture of microlayer, and relate those (quantitatively) to fluid properties and the rupture condition (the critical heat flux). 3. Determine the extent of applicability of scales separation in reduced gravity; that is, the point where CHF becomes water-supply-limited. 4. Determine the infleunce of convection on the results of Objetives 1 and 2. 5. Translate the results of all previous objectives to terms usable for predicting and enhancing CHF, as well as for use in a two-phase CFD framework. This project will provide a basis to develop a physics-based understanding of Boiling Crisis phenomenon in pool boiling (horizontal and vertical flat plates), and weakly-convecting flow boiling (vertical channel flow). We can then translate this understanding to a means for quantitative prediction of critical heat flux, as a function of local (near-wall) two-phase flow conditions, appropriate and convenient for interrogation (modelling) in the frame of detailed multiphase CFD simulations. Furthermore, the thus-gained understanding and experience can be used to enhance the coolability limits in both pool and forced convection flows by manipulating surface characteristics, and perhaps even flow geometries.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has been leveraged with a contract from the Department of Energy concerning the coolability limits of nuclear reactor vessels that in case of severe accidents are immersed in water pools by flooding the outside area with water. In turn the results of this fundamentally oriented work had a tremendous influence and impact on the large scale experiments we run for this nuclear application, and helped resolve this practical issue even though we are still in the process of developing a basic understanding.
Task Progress:	The work in this project capitalizes on advanced experimental and diagnostic techniques that have been developed at UCSB, namely the use of high-speed, high-resolution infrared thermometry on nano-film heaters that have been vapor-deposited on thin glass substrates. During the past year, the project focus was placed on mechanisms which govern Microlayer rupture in pool boiling. We have completed an investigation of the effect of water chemistry, surface treatment, and heater material on pool boiling CHF [10,11]. Extensive measurements were performed with X-Ray radiography to further quantify the scale separation phenomenon discovered in our previous study on BETA facility [1,2]. Theoretical research was directed toward understanding mechanisms that control heterogeneous nucleation, including that on nanoscopically smooth surfaces [5-7,12]. The study indicated that both surface heterogeneity (Franken islands) and gas solubility in water may have contributed to the reduced nucleation energy barrier. Boiling Crisis (BC) on horizontal, upwards-facing copper and steel surfaces under the influence of various coolant chemistries was investigated [10,11]. In addition to Boric Acid (BA) and TriSodium Phosphate (TSP), pure De-Ionized Water (DIW) and Tap Water (TW) are included in experiments carried out in the BETA facility. We found that the presence of chemicals TSP and BA enhances the coolability of fresh surfaces well beyond even those that have been fully-aged. The so-enhanced fluxes, reaching up to ~ 1.9 MW/m2 motivate examination of such effects in large-scale, convective-boiling situations. Furthermore, the effect of coolant chemistry through molecular scale liquid-solid interactions at the contact line gives another new evidence to further support the idea (Theofanous et al, 2002 [1,2]) that the basic mechanism of burnout is microlayer rupture by contact line instability. We have conducted a series of experiments on nanofluid boiling and nanofluid burnout [6, 7]. We discovered that presence of nanoparticles in the coolant alters both nucleation and nucleate boiling heat transfer behavior, making a fresh heater to behave like an aged heater and significantly improving the heater’s resilience to burnout. We measured contact line motions for clean water and nanofluid and investigated mechanisms by which non-hydrodynamic factors govern the burnout [2,5-7]. A basic understanding derived from our experiments and theoretical analysis led us to suggest that molecular wicking and nanoscale forces as manifested in the effect of coolant chemistry and nanoparticles offer new avenues to control and enhance resistance to burnout in pool boiling as well as in flow boiling and spray cooling. This work will also elucidate film dryout, i.e. burnout in high quality boiling as is of current interest to NASA for the nuclear power Rankin cycle effort. Finally, we have developed and passed through preliminary testing one new experimental apparatus as follows: Synchronization high speed IR (recently update maximum speed to 20000 fps), a new high speed video camera (a phantom V7.0, borrowed from one of our other project, the maximum speed up to 160,000 fps) and 100 mW green laser to discover the scales separation phenomenon. We are successful to produce a stable ultra-thin water film. Without disturbing the film, and yet at very high magnification as needed not only to “see” microscope features of the film and contact line motions, but also to know the temperature affected by those phenomenon. References: 1. T.G. Theofanous, J.P. Tu, A.T. Dinh and T.N. Dinh, The Boiling Crisis Phenomenon. Part 1: Nucleation and Nucleate Boiling Heat Transfer, J. Experimental Thermal and Fluid Science, pp.775-792, V.26 (6-7) August 2002. 2. T.G. Theofanous, T.N. Dinh, J.P. Tu, and A.T. Dinh, The Boiling Crisis Phenomenon. Part 2: Dryout Dynamics and

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13. T.G. Theofanous, T.N. Dinh, “High heat flux boiling and burnout as microphysical phenomena: Mounting evidence and opportunites”, The 7th Japan-US Seminar on Two-Phase Flow Dynamics, Nagahama, Shiga, December 6-11, 2004.

Bibliography Type:	Description: (Last Updated: 10/17/2007)
Papers from Meeting Proceedings	Theofanous TG, Dinh TN. "High heat flux boiling and burnout as microphysical phenomena: mounting evidence and opportunities." 7th Japan-US Seminar on Two-Phase Flow Dynamics, Nagahama , Shiga, December 7-11, 2004. Proceedings, 7th Japan-US Seminar on Two-Phase Flow Dynamics, December 2004. , Dec-2004